(ii) Let X and Y be two sets. Let c be a fixed element of Y. The function $f : X \to Y$ defined by $f(x) = c$ for all $x \in X$ is called a ***constant function*** (See Figure 1.20).

The value of a constant function is same for all values of x throughout the domain.

If X and Y are $\mathbb{R}$, then the graph of the identity function and a constant function are as in Figures 1.21 and 1.22. If X is any set, then the constant function defined by $f(x) = 0$ for all x is called the ***zero function***. So zero function is a particular case of constant function.

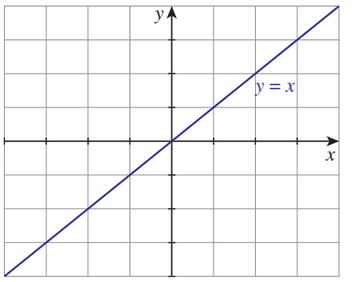


Figure 1.21

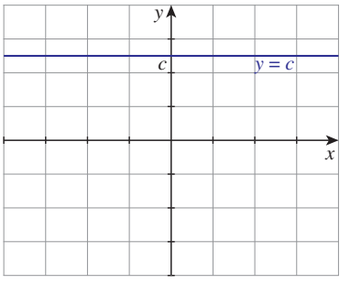


Figure 1.22

(iii) The function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = |x|$, where $|x|$ is the modulus or absolute value of x, is called the ***modulus function*** or ***absolute value function***. (See Figure 1.23.)

Let us recall that $|x|$ is defined as

$$|x| = \begin{cases} -x & \text{if} \quad x < 0 \\ 0 & \text{if} \quad x = 0 \\ x & \text{if} \quad x > 0 \end{cases} \quad \text{or} \quad |x| = \begin{cases} -x & \text{if} \quad x \leq 0 \\ x & \text{if} \quad x > 0 \end{cases} \quad \text{or} \quad |x| = \begin{cases} -x & \text{if} \quad x < 0 \\ x & \text{if} \quad x \geq 0 \end{cases}$$

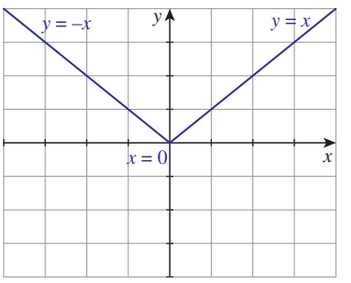


Figure 1.23

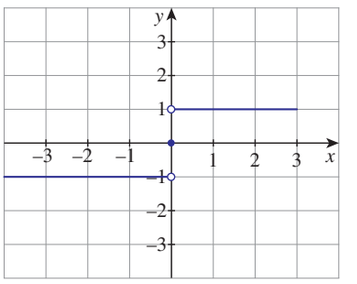


Figure 1.24

(iv) The function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = \begin{cases} \frac{x}{|x|} & \text{if} \quad x \neq 0 \\ 0 & \text{if} \quad x = 0 \end{cases}$ is called the ***signum function***. This function is denoted by sgn. (See Figure 1.24)

(v) The function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x)$ is the greatest integer less than or equal to x is called the integral part function or the ***greatest integer function*** or the ***floor function***. This function is denoted by $\lfloor x \rfloor$. (See Figure 1.25.)

(vi) The function $f : \mathbb{R} \to \mathbb{R}$defined by $f(x)$ is the smallest integer greater than or equal to x is called the ***smallest integer function*** or the ***ceil function*** (See Figure 1.26.). This function is denoted by $\lceil \cdot \rceil$; that is $f(x)$ is denoted by $\lceil x \rceil$.

The functions (v) and (vi) are also called ***step functions***.

Let us note that $\lfloor 1\frac{1}{5} \rfloor = 1, \lfloor 7.23 \rfloor = 7, \lfloor -2\frac{1}{2} \rfloor = -3$ (not -2), $\lfloor 6 \rfloor = 6$ and $\lfloor -4 \rfloor = -4$.

Let us note that $\lceil 1\frac{1}{5} \rceil = 2, \lceil 7.23 \rceil = 8, \lceil -2\frac{1}{2} \rceil = -2$ (not -3), $\lceil 6 \rceil = 6$ and $\lceil -4 \rceil = -4$.